

# Comparing Proportions

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This page reviews tests for comparing groups on a categorical outcome: the chi-square test and Fisher's exact test for contingency tables. It uses the chi-square reference distribution defined on the [Statistical Inference](#) page. This page is adapted from Vittinghoff et al. (2012), Chapter 3.

## The HERS data

The examples on this page use the HERS data, which the [Comparing Means](#) page describes. The `rmb` R package includes the dataset; `haven::as_factor()` converts its Stata value labels to factors:

```
hers <- rmb::hers |> haven::as_factor()
```

## Comparing two groups: categorical outcomes

### Contingency tables

**Example 0.1** (Exercise by treatment group in HERS). Table 1 cross-tabulates regular exercise at baseline by treatment group, as a [contingency table](#) with row percentages.

Table 1: Exercise by treatment group in HERS

```
hers |>
  gtsummary::tbl_cross(
    row = exercise,
    col = HT,
    label = list(
      exercise ~ "Exercises regularly",
      HT ~ "Treatment group"
    ),
    percent = "row"
  )
```

|                     | Treatment group |                 | Total        |
|---------------------|-----------------|-----------------|--------------|
|                     | placebo         | hormone therapy |              |
| Exercises regularly |                 |                 |              |
| no                  | 853 (50%)       | 842 (50%)       | 1,695 (100%) |
| yes                 | 530 (50%)       | 538 (50%)       | 1,068 (100%) |
| Total               | 1,383 (50%)     | 1,380 (50%)     | 2,763 (100%) |

## The chi-square test

**Definition 0.1** (Expected count under independence). Let a contingency table have  $r$  rows and  $c$  columns, with observed count  $O_{ij}$  in row  $i$  and column  $j$ , row totals  $R_i$ , column totals  $C_j$ , and grand total  $n$ . The **expected count** in cell  $(i, j)$  under independence is

$$E_{ij} \stackrel{\text{def}}{=} \frac{R_i C_j}{n}.$$

**Example 0.2** (Expected count in a 2 x 2 table). In a table with  $n = 100$ , first-row total  $R_1 = 30$ , and first-column total  $C_1 = 40$ , the expected count in cell  $(1, 1)$  is  $E_{11} = 30 \cdot 40 / 100 = 12$ .

**Definition 0.2** (Pearson's chi-square test of independence). With observed counts  $O_{ij}$  and **expected counts**  $E_{ij}$  in an  $r \times c$  contingency table, **Pearson's chi-square test** of the null hypothesis that the row and column variables are independent uses the statistic

$$X^2 \stackrel{\text{def}}{=} \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Its p-value is  $\Pr(W \geq X^2)$ , where  $W$  has the  $\chi_{(r-1)(c-1)}^2$  distribution ([chi-square distribution](#)).

**Theorem 0.1** (Large-sample null distribution of the chi-square statistic). *Let  $n$  observations be sampled independently and classified by two categorical variables, and let the two variables be independent. Then as  $n \rightarrow \infty$ , the distribution of  $X^2$  (Definition 0.2) converges to the  $\chi_{(r-1)(c-1)}^2$  distribution (Hogg et al. 2015).*

The chi-square approximation is poor when some expected counts are small. A common rule of thumb asks for every  $E_{ij}$  to be at least 5.

**Example 0.3** (Chi-square test of exercise by treatment group in HERS). The expected counts and statistic of Definition 0.2 for Table 1:

```
observed <- table(hers$exercise, hers$HT)
expected <- outer(rowSums(observed), colSums(observed)) / sum(observed)
x_squared <- sum((observed - expected)^2 / expected)
expected
#>      placebo hormone therapy
#> no      848.42      846.58
#> yes     534.58      533.42
c(
  x_squared = x_squared,
  p_value = pchisq(x_squared, df = 1, lower.tail = FALSE)
)
#> x_squared  p_value
#> 0.128054  0.720458
```

Every expected count is large, so the  $\chi_1^2$  approximation of Theorem 0.1 is reasonable. `chisq.test()` with `correct = FALSE` reports the same values:

```
chisq.test(hers$exercise, hers$HT, correct = FALSE)
#>
#> Pearson's Chi-squared test
#>
#> data:  hers$exercise and hers$HT
#> X-squared = 0.1281, df = 1, p-value = 0.72
```

The p-value is large: the data give no evidence that exercise depends on treatment group, as randomization would lead us to expect.

For a  $2 \times 2$  table, `chisq.test()` applies Yates' continuity correction by default, which subtracts 0.5 from each  $|O_{ij} - E_{ij}|$  before squaring, and so gives a smaller statistic than Definition 0.2. `correct = FALSE` turns the correction off.

### Fisher's exact test

**Definition 0.3** (Fisher's exact test). Take a  $2 \times 2$  contingency table with cells  $a$ ,  $b$ ,  $c$ , and  $d$  as in [the contingency table definition](#), and hold its row and column totals fixed. Under independence of the row and column variables, the probability that the top-left cell equals  $x$  is the hypergeometric probability

$$p(x) \stackrel{\text{def}}{=} \frac{\binom{a+b}{x} \binom{c+d}{a+c-x}}{\binom{n}{a+c}}.$$

**Fisher's exact test** of independence has p-value

$$\sum_{x: p(x) \leq p(a)} p(x),$$

the total probability of the tables with the same totals that are no more probable than the observed table.

The p-value is exact: it comes from the null distribution itself, not from a large-sample approximation, so the test is valid even when expected counts are small, and it is often used for  $2 \times 2$  tables in which some expected count is below 5 (Theorem 0.1). Other two-sided versions exist; this one is the version that R's `fisher.test()` computes.

**Example 0.4** (Fisher's exact test of exercise by treatment group in HERS). The p-value of Definition 0.3 for Table 1, computed from the hypergeometric probabilities with `dhyper()`:

```

a <- observed[1, 1]
row1 <- sum(observed[1, ])
row2 <- sum(observed[2, ])
col1 <- sum(observed[, 1])
x <- max(0, col1 - row2):min(row1, col1)
p_x <- dhyper(x, m = row1, n = row2, k = col1)
p_a <- dhyper(a, m = row1, n = row2, k = col1)
sum(p_x[p_x <= p_a * (1 + 1e-7)])
#> [1] 0.72528

```

The tolerance `1e-7` keeps tables whose probability equals  $p(a)$  up to rounding error, as `fisher.test()` does. `fisher.test()` reports the same p-value:

```

fisher.test(hers$exercise, hers$HT)
#>
#> Fisher's Exact Test for Count Data
#>
#> data: hers$exercise and hers$HT
#> p-value = 0.725
#> alternative hypothesis: true odds ratio is not equal to 1
#> 95 percent confidence interval:
#> 0.879664 1.202192
#> sample estimates:
#> odds ratio
#> 1.02836

```

With counts this large, the exact p-value is close to the chi-square p-value of Example [0.3](#).

## Measures of association for $2 \times 2$ tables

Tests of independence say whether two binary variables are associated, but not how strongly. Risk differences, risk ratios, and odds ratios measure the strength of the association; see [Odds Ratios and Relative Risks](#).

## References

Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman. 2015. *Probability and Statistical Inference*. Ninth edition. Pearson.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.